

SISTEM INFORMASI SMA CENDEKIA NUSANTARA BERBASIS SCALABLE WEB APPLICATION

*Disusun untuk Memenuhi Tugas Akhir Mata Kuliah Advanced Software Testing
and Quality Assurance*

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FAKULTAS TEKNIK

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DAFTAR ISI

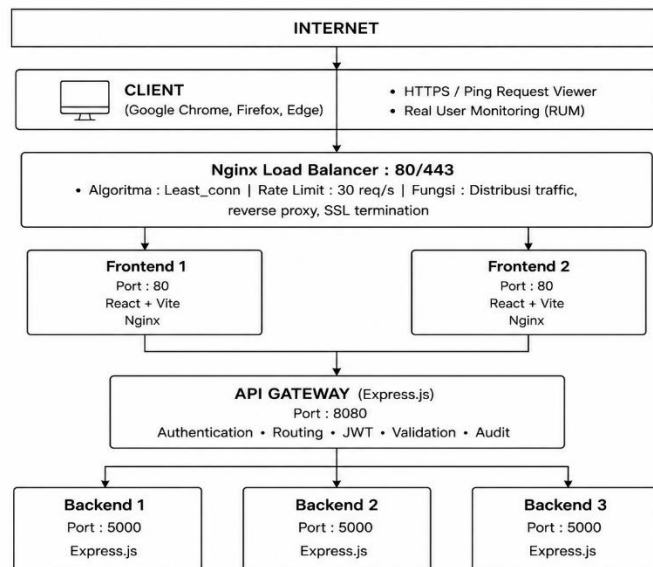
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1. Pendahuluan

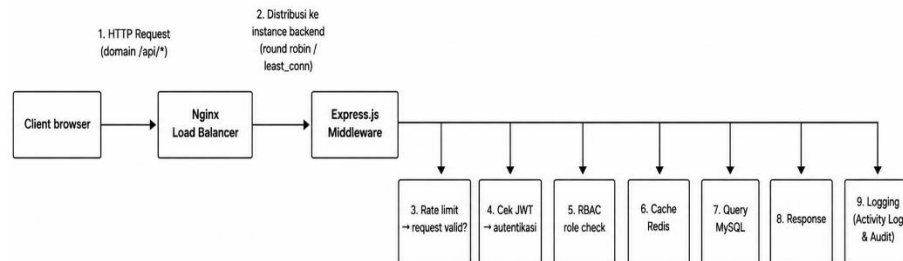
Dokumen Software Design Document (SDD) ini mendeskripsikan desain teknis sistem Informasi SMA Cendekia Nusantara secara komprehensif, meliputi arsitektur tingkat tinggi, struktur modul, API contract, dan skema database. Dokumen ini berfungsi sebagai referensi teknis untuk tim QA dalam menyusun test plan, memahami alur data antar modul, dan menentukan integration test scenarios.

2. High-Level Architecture Diagram

2.1 Gambaran Arsitektur Keseluruhan



2.2 Alur Request Detail



2.3 Strategi Scaling

Strategi	Mekanisme	Implementasi
Horizontal Scaling	Tambah instance backend	docker-compose scale backend=3 (Stateless JWT, tanpa ubah kode)
Vertical Scaling	Tambah resource container	Sesuaikan CPU dan mem_limit di docker-compose.yml
Caching (Redis)	In-memory cache data statis	GET /api/teachers, /api/students, /api/subjects di-cache 5–10 menit
Connection Pooling	Prisma pool MySQL	Max 10 connections per instance, mencegah exhaustion
Load Balancing	Nginx least_conn	Request diarahkan ke backend dengan koneksi aktif paling sedikit

2.4 Rancangan Virtual Server (8 vCPU)

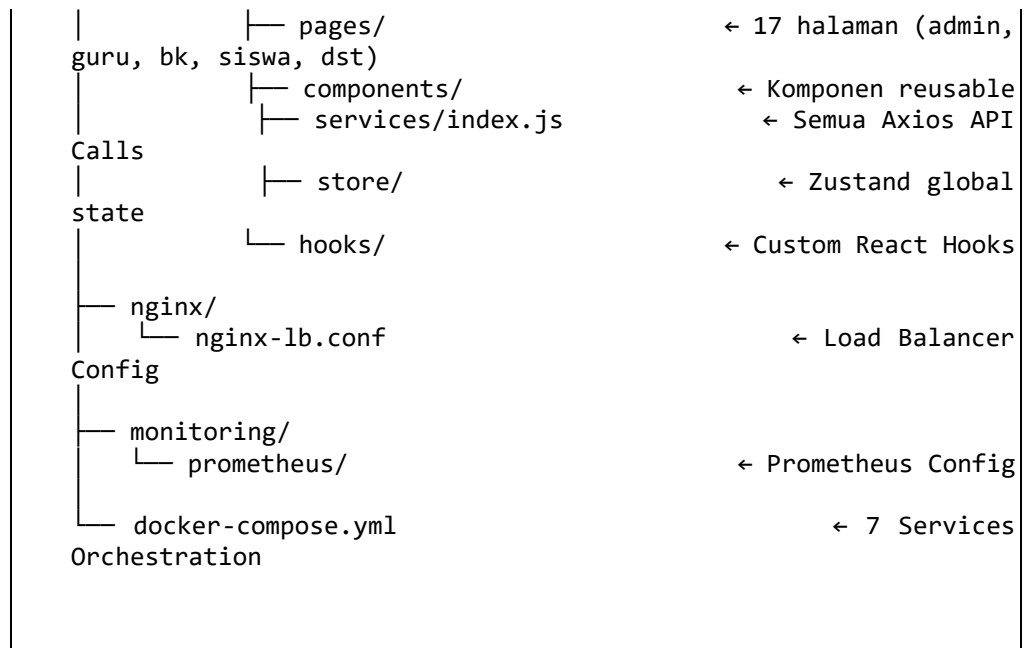
vCPU	Service	Port	Alasan Pemisahan
1	Frontend (React + Nginx)	80	Serve static files; dapat diganti CDN di produksi
2	API Gateway + Nginx LB	8080	Entry point semua traffic, rate limiting, SSL termination
3	Modul Jurnal	5000	High write frequency; semua guru input jurnal bersamaan
4	Modul BK	5000	Data sensitif, butuh isolasi dari modul lain
5	Modul Kesiswaan	5000	Bulk operation (import 100+ siswa), export Excel intensif CPU
6	MySQL Database	3306	Dedicated CPU agar query tidak bersaing dengan aplikasi
7	Redis Cache	6379	In-memory; butuh CPU dedicated untuk latensi konsisten
8	Prometheus + Grafana	9090/3000	Monitoring dipisah agar tidak mengganggu sistem utama

3. Detail Modul & Structural Design

3.1 Struktur Direktori Proyek

<pre>sisekolah/ ├── backend/ │ ├── src/ │ │ ├── server.js │ │ └── global middleware, audit │ └── config/</pre>	← Entry point,
--	----------------

- database.js	← Prisma client
- singleton - redis.js	← Redis client +
- cache.get/set/del/flush - logger.js	← Winston logger
(file + console) - middlewares/ - auth.middleware.js	← authenticate() +
- authorize(roles) - logger.middleware.js	← Request audit log
- writer - modules/ - auth/	← login, logout,
- refresh, changePassword - users/	← CRUD user + RBAC
- management - students/	← CRUD siswa,
- export/import Excel - teachers/	← CRUD guru, cache
- Redis - classes/	← CRUD kelas +
- penugasan wali kelas - subjects/	← CRUD mata pelajaran
- grades/	← Input nilai, rekap
- per semester - attendance/	← Input absensi
- bulk, rekap - journals/	← Jurnal harian guru
- bk/	← Kasus BK,
- pelanggaran, prestasi - schedules/	← Jadwal pelajaran
- dashboard/	← Statistik +
- activity logs - reports/	← Laporan + active-
- semester - services/	
- metrics.js	← Prometheus prom-
- client metrics - utils/	
- response.js	← successResponse,
- errorResponse, pagination - prisma/	
- schema.prisma	← 22 tabel, 33
- relasi, 26 index - seeders/	← Data awal: 100
- siswa, 20 guru, 10 kelas - frontend/	
- src/	

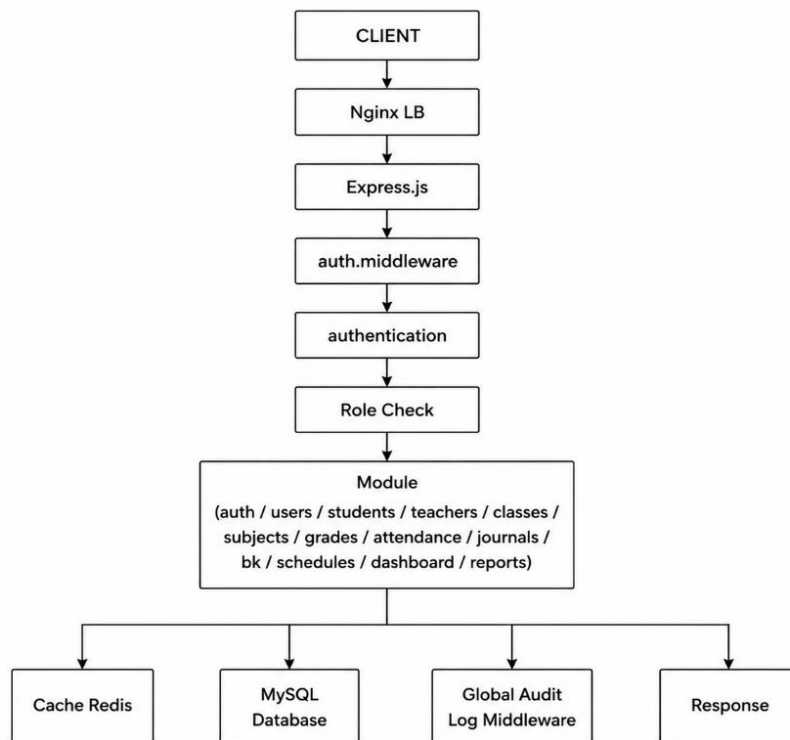


3.2 Deskripsi Modul Backend

Modul	Tanggung Jawab	Cache Redis	Dependensi
auth	Login, JWT generation, refresh token, logout, ganti password	Tidak	users, roles, refresh_tokens
users	CRUD pengguna sistem, penugasan role	Tidak	roles
students	CRUD siswa, import/export Excel, mutasi siswa	Tidak (read cache)	users, classes, grades, attendance, bk
teachers	CRUD guru, penugasan mengajar, cache invalidation	Ya (teachers:*)	users, subjects, classes
classes	CRUD kelas, penugasan wali kelas, assign siswa	Tidak	teachers, academic_years

Modul	Tanggung Jawab	Cache Redis	Dependensi
subjects	CRUD mata pelajaran	Ya (subjects:all)	grades, schedules
grades	Input nilai tugas/UTS/UAS, hitung nilai akhir, export	Tidak	students, subjects, semesters
attendance	Input absensi bulk (deleteMany+createMany), rekap statistik	Tidak	students, classes, semesters
journals	Jurnal harian guru, CRUD dengan validasi teacherId	Tidak	teachers, classes, subjects
bk	Kasus konseling, pelanggaran, prestasi, catatan BK	Tidak	students, teachers (counselor)
schedules	Jadwal pelajaran per kelas per semester	Ya (schedules:*)	classes, subjects, teachers, semesters
dashboard	Statistik sistem, activity logs (audit trail)	Sebagian	Semua modul
reports	Laporan nilai/absensi, active- semester, academic-years	Tidak	grades, attendance, semesters

3.3 Alur Interaksi Antar Modul



Contoh Alur: POST /api/attendance (Input Absensi Bulk)

1. Frontend kirim POST /api/attendance dengan JWT token
2. `authenticate()` → verifikasi token → ambil user dari DB → `req.user = user`
3. `authorize(['guru', 'wali_kelas', 'admin'])` → cek role
4. Handler attendance: terima array data absensi
5. `prisma.$transaction([`
 - `deleteMany({ classId, date, semesterId })),` ← hapus absensi lama
 - `createMany({ data: attendanceRecords })` ← buat absensi baru`])`

```
6. Audit log: action=CREATE_ATTENDANCE, module=attendance
7. successResponse(res, result, 'Absensi berhasil disimpan', 201)
```

4. API Contract

4.1 Autentikasi

Method	Endpoint	Auth	Request Body	Response
POST	/api/auth/login	False	{ identifier, password }	200: { accessToken, refreshToken, user } atau 401: Login gagal
POST	/api/auth/refresh	False	{ refreshToken }	200: { accessToken } atau 401: Token invalid
POST	/api/auth/logout	True	—	200: Logout berhasil
GET	/api/auth/me	True	—	200: { user (id, student, teacher) }
PUT	/api/auth/change- password	True	{ oldPassword, newPassword }	200: Berhasil atau 400: Password lama salah

4.2 Manajemen Siswa

Method	Endpoint	Role	Query/Body	Response
GET	/api/students	Semua	page=1&limit=10&search=&classId=&status=	200: Paginated list siswa
GET	/api/students/:id	Semua	—	200: Detail siswa + relasi
POST	/api/students	admin, wali_kelas	{ fullName, nis, nisn, gender, birthPlace, birthDate, address,	201: Siswa dibuat atau 409: NIS/NISN duplikat

Method	Endpoint	Role	Query/Body	Response
			phone, religion, entryYear, classId, email }	

4.3 Manajemen Siswa (Lanjutan)

Method	Endpoint	Role	Query/Body	Response
PUT	/api/students/:id	admin, wali_kelas	{ fullName?, gender?, address?, phone?, status }	200: Data diperbarui
DELETE	/api/students/:id	admin	—	200: Berhasil dihapus (cascade)
GET	/api/students/:id/grades	Semua	—	200: Nilai siswa per semester
GET	/api/students/:id/attendance	Semua	—	200: Absensi summary + statistik
POST	/api/students/import/excel	admin, wali_kelas	multipart/form-data file (.xlsx)	200: Hasil import (success, skipped, error)
GET	/api/students/export/excel	admin, wali_kelas	—	200: File .xlsx

4.4 Absensi

Method	Endpoint	Role	Body/Query	Response
GET	/api/attendance	Semua	?classId&date&semesterId&page&limit	200: List absensi (paginated)
POST	/api/attendance	guru, wali_kelas, admin	{ classId, date, semesterId, records: [{ studentId, status, note }] }	201: Absensi disimpan (deleteMany+createMany)
GET	/api/attendance/summary	Semua	?classId&semesterId	200: Rekap statistik absensi per siswa

4.5 Nilai (Grades)

Method	Endpoint	Role	Body/Query	Response
GET	/api/grades	Semua	?classId&subjectId&semesterId&page&limit	200: List nilai (paginated)
POST	/api/grades	guru, wali_kelas, admin	{ studentId, subjectId, semesterId, tugas, uts, uas, finalScore }	201: Nilai disimpan
PUT	/api/grades/:id	guru, wali_kelas, admin	{ tugas?, uts?, uas?, finalScore? }	200: Nilai diperbarui
DELETE	/api/grades/:id	admin	—	200: Nilai dihapus

4.6 BK & Jurnal

Method	Endpoint	Role	Body/Query	Response
GET	/api/bk/cases	Semua (terbatas)	?studentId&status&page&limit	200: List kasus BK
POST	/api/bk/cases	guru_bk, admin	{ studentId, counselorId, title, description, type, status }	201: Kasus dibuat
GET	/api/bk/violations	Semua (terbatas)	?studentId&page&limit	200: List pelanggaran
POST	/api/bk/violations	guru_bk, wali_kelas, admin	{ studentId, type, description, date, points }	201: Pelanggaran dicatat
GET	/api/bk/achievements	Semua	?studentId&page&limit	200: List prestasi
POST	/api/bk/achievements	guru_bk, wali_kelas, admin	{ studentId, title, category, level, date, description }	201: Prestasi dicatat
GET	/api/journals	Semua (terbatas)	?teacherId&classId&page&limit	200: List jurnal
POST	/api/journals	guru, wali_kelas, admin	{ teacherId, classId, subjectId, date, material, activity, notes }	201: Jurnal dicatat

4.7 Dashboard & Laporan

Method	Endpoint	Role	Query	Response
GET	/api/dashboard/stats	admin, kepek	—	200: Statistik sistem (total siswa, guru, kelas, dsb)
GET	/api/dashboard/student/:id	Semua	—	200: Dashboard personal siswa
GET	/api/dashboard/activity-logs	admin	?module&page&limit	200: Audit log (paginated)
GET	/api/reports/active-semester	Semua	—	200: Semester aktif
GET	/api/reports/academic-years	Semua	—	200: List tahun ajaran
GET	/api/reports/attendance/:semesterId	admin, kepek	?classId	200: Laporan absensi per semester
GET	/api/reports/grades/:semesterId	admin, kepek	?classId&subjectId	200: Laporan nilai per semester
GET	/health	No (public)	—	200: { status: "healthy", uptime, version }
GET	/metrics	No (public)	—	200: Prometheus metrics format

5. (Database Schema/ERD)

5.1 Daftar Tabel

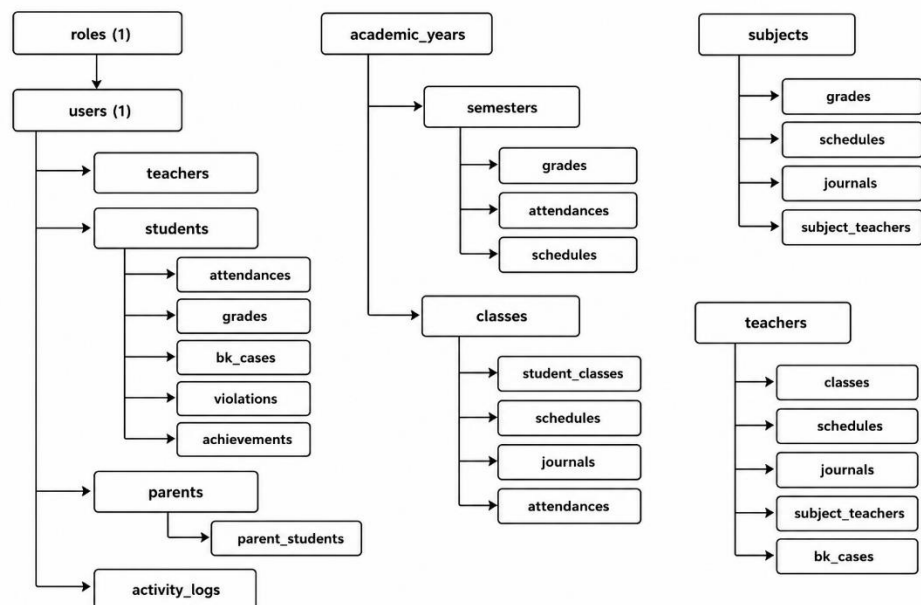
No	Nama Tabel	Deskripsi	Kolom Utama	Relasi
1	roles	Peran pengguna sistem	id, name, description	→ users (1:N)
2	users	Akun login pengguna	id, username, email, password, roleId, isActive, lastLogin	→ teacher/student/parent (1:1)
3	refresh_tokens	JWT refresh token storage	id, userId, token, expiresAt	← users (N:1)
4	teachers	Data guru	id, userId, nip, fullName, gender, birthDate, education, position	← users; → classes, subjects, journals
5	students	Data siswa	id, userId, nis, nisn, fullName, gender, birthDate, status, entryYear	← users; → classes, grades, attendance, bk
6	parents	Data orang tua/wali	id, userId, fullName, phone, job, address	← users; ↔ students (N:M via parent_students)

No	Nama Tabel	Deskripsi	Kolom Utama	Relasi
7	parent_students	Relasi orang tua - siswa	id, parentId, studentId, relation	Join table
8	academic_years	Tahun ajaran	id, name, startDate, endDate, isActive	→ classes, semesters
9	semesters	Semester per tahun ajaran	id, yearId, name (Ganjil/Genap), startDate, endDate, isActive	← academic_years; → grades, attendance
10	classes	Data kelas	id, name, grade (X/XI/XII), yearId, homeroomId	← teachers (homeroom); → student_classes, schedules
11	student_classes	Relasi siswa - kelas	id, studentId, classId	Join table
12	subjects	Mata pelajaran	id, code, name, creditHours, description	→ grades, schedules, subject_teachers
13	subject_teachers	Relasi guru - mata pelajaran	id, teacherId, subjectId	Join table

No	Nama Tabel	Deskripsi	Kolom Utama	Relasi
14	grades	Nilai akademik siswa	id, studentId, subjectId, semesterId, tugas, uts, uas, finalScore	← students, subjects, semesters
15	attendances	Absensi harian siswa	id, studentId, classId, semesterId, date, status (hadir/sakit/izin/alpha), note	← students, classes, semesters
16	schedules	Jadwal pelajaran	id, classId, subjectId, teacherId, semesterId, day, startTime, endTime	← classes, subjects, teachers, semesters
17	journals	Jurnal mengajar harian	id, teacherId, classId, subjectId, date, material, activity, notes	← teachers, classes, subjects
18	bk_cases	Kasus konseling BK	id, studentId, counselorId, title, description, type, status, date	← students, teachers
19	bk_notes	Catatan tindak lanjut kasus BK	id, caseId, content, date, nextDate	← bk_cases
20	violations	Pelanggaran disiplin siswa	id, studentId, type, description, date, points, reportedBy	← students
21	achievements	Prestasi siswa	id, studentId, title, category, level, date, description	← students

No	Nama Tabel	Deskripsi	Kolom Utama	Relasi
22	activity_logs	Audit trail operasi sistem	id, userId, action, module, description, ipAddress, userAgent, createdAt	← users

5.2 Diagram Relasi (Ringkasan)



5.3 Index & Optimasi

Tabel	Indeks	Tujuan Optimasi
users	username, email (UNIQUE)	Login lookup O(1)
students	nis (UNIQUE), nisan (UNIQUE), fullName, status	Search dan filter siswa
teachers	nip (UNIQUE), fullName	Search guru

Tabel	Indeks	Tujuan Optimasi
grades	studentId, subjectId, semesterId (composite)	Query nilai per semester
attendances	studentId, classId, date, semesterId (composite)	Query absensi bulk dan rekap
activity_logs	userId, module, createdAt	Filter audit log
refresh_tokens	token (UNIQUE), userId	Token lookup saat refresh
schedules	classId, semesterId, teacherId, day	Query jadwal per kelas/hari
bk_cases	studentId, status	Filter kasus aktif per siswa
violations	studentId, date	Riwayat pelanggaran per siswa

Total Database: 22 tabel | 33 relasi | 26 index

Database Engine: MySQL 8.0 dengan InnoDB (mendukung foreign key constraints dan ACID transactions)

ORM: Prisma 5.x dengan type-safe queries dan migration system

Transaction: Digunakan pada operasi multi-tabel: create siswa (user+student+studentClass), delete siswa (cascade 8 tabel), input absensi bulk (deleteMany+createMany)

